

# Muhammad Ekramah

Bioinformatics Graduate | Genomics and Computational Biology

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## Profile

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Bioinformatics graduate from Quaid-i-Azam University with a background in genomics, biostatistics and computational biology. My academic projects have involved machine learning for antimicrobial-resistance prediction, RNA-seq analysis and biological software development. Through these projects and my internship at the National Institutes of Health, I developed a particular interest in understanding how computational findings relate to biological processes. I aim to strengthen my statistical and computational skills to investigate complex genomic data and biological systems.

## Education

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### BS in Bioinformatics

Sep 2022 – Sep 2026

National Centre for Bioinformatics, Quaid-i-Azam University, Islamabad, Pakistan

CGPA: 3.4/4.0

*Relevant coursework:* Probability and Biostatistics; Bioinformatics Computing; Genomics; Functional Genomics; Systems Biology; Artificial Intelligence; Data Structures and Algorithms; Modelling and Simulation; Database Management Systems.

## Selected Academic Projects

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### Antimicrobial-resistance prediction in *Neisseria gonorrhoeae*

Repository

*Bioinformatics Computing II* | 2025–2026

- Worked with genomic unitigs and 3,786 metadata records, matching labelled isolates to antibiotic-specific features for azithromycin and ciprofloxacin resistance prediction.
- Compared logistic regression and random forest with a prevalence baseline. Used five-fold cross-validation grouped by identical genomic profiles, with variance filtering and chi-square feature selection within training folds.
- Examined class imbalance, bootstrap confidence intervals and geographic differences in model performance. Logistic regression achieved internal balanced accuracy of 0.949 for azithromycin and 0.969 for ciprofloxacin.
- Used BLAST to examine selected unitigs, including alignments to regions associated with *penA* and *gyrA*. Since the supplied unitigs had already been GWAS-filtered, the predictive results still need independent validation.

### RNA-seq analysis of Notch2 knockout neural stem cells

Repository

*Genomics project* | 2025–2026

- Analysed the public GSE116773 dataset to compare Notch2 conditional-knockout and wild-type mouse hippocampal neural stem cells. Combined 24 technical count files into six biological RNA preparations, three per condition.
- Used PyDESeq2, negative-binomial modelling and Benjamini–Hochberg correction. Identified 121 genes at FDR < 0.05 among 21,531 retained genes (77 higher and 44 lower in the knockout samples).
- Explored sample differences using PCA, sample-distance plots and gene expression plots. WT3 differed from the other samples, an important consideration given the small number of biological replicates.

## Selected Academic Projects (continued)

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**PathoScope AI: viral genomics and transcriptomics prototype**     [Repository](#)   |   [Video demo](#)  
*Functional Genomics team project / Mar–Aug 2026*

- Worked with a team on a web-based prototype using FastAPI, PostgreSQL and a Next.js interface. My contributions focused on workflow execution, FASTQ integration, PubMed retrieval and report generation.
- Worked on sequential pipeline steps, job-status tracking, database connectivity and literature retrieval using NCBI E-Utilities. The team integrated read-processing tools including FastQC, fastp and SPAdes.
- The prototype also includes ORF scanning, protein annotation tools and an exploratory gene-expression module. Some fallback functions use simulated results; the software is not a validated diagnostic pipeline.

## Experience

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**Research Intern** *Jul–Aug 2025*  
National Institutes of Health (NIH), Islamabad, Pakistan

- Used R and the AMR package to analyse antimicrobial-resistance surveillance data, including wastewater data, and summarise susceptibility patterns.
- Gained exposure to PCR-based diagnostics, Oxford Nanopore MinION sequencing and bulk RNA-seq workflows.

**Python Developer Intern** *Jul–Sep 2024*  
Digital Empowerment Network | Remote

- Worked on Python backend services and REST APIs using FastAPI and PostgreSQL.
- Automated data cleaning and processing with Pandas and NumPy, and developed visualisation utilities for GitHub Classroom analytics as part of a team.

## Technical Skills

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|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming</b>         | Python (BioPython , Pandas, NumPy, scikit-learn), R, Bash, SQL                                                                                       |
| <b>Statistics and data</b> | Classification; grouped cross-validation; feature selection; bootstrap intervals; PCA; differential-expression analysis; multiple-testing correction |
| <b>Bioinformatics</b>      | Genomic unitigs; bulk RNA-seq; BLAST; PyDESeq2; DESeq2; FastQC, fastp, HISAT2 and featureCounts                                                      |
| <b>Software</b>            | Git/GitHub; Jupyter; FastAPI; PostgreSQL; Linux                                                                                                      |
| <b>Data resources</b>      | NCBI GEO/SRA; PubMed; Ensembl; UniProt and Pfam (team-project use)                                                                                   |
| <b>Wet-lab exposure</b>    | DNA extraction , PCR-based diagnostics; Oxford Nanopore MinION sequencing; familiarity with molecular biology laboratory workflows                   |

## Research Interests

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Understanding how genetic variation in bacteria contributes to antimicrobial resistance and how reliably genomic prediction models perform across different populations. I am also interested in studying environmental microbial communities through metagenomics and using transcriptomic data to investigate gene-expression changes and their underlying biological pathways.